VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI-590 018, KARNATAKA

Synopsis
on

"FAKE PRODUCT IDENTIFICATION BY QR USING BLOCKCHAIN"

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Project Associates

DISHA D PUJAR	**2SR22CS012**
KAVYA R GYANAGOUDRA	**2SR22CS019**
NAMRATHA V KENCHARADDI	**2SR22CS027**
VANAJA H MALLUR	**2SR22CS052**

Under the Guidance of:
Dr. Murgesh V Jambigi **Ph.D**
Associate Prof., Dept. of CS&E

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SRI TARALABALU JAGADGURU INSTITUTE OF TECHNOLOGY
RANEBENNUR-581 115

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INTRODUCTION

Each day brings more technological progress. Today's company market is competitive, and everyone strives to give customers the best options. Some people sell unlicensed things to get rich quickly. The Organisation for Economic Co-operation and Development (OECD) and the European Union Intellectual Property Office (2019) define counterfeit goods as products that copy the appearance of genuine goods without permission. Recent years have seen a boom in counterfeit goods, and the spectrum of products in danger has grown. With almost half of US Customs and Border Protection's counterfeit items being clothes and accessories, copyright violations have increased. While international trade has remained flat, counterfeit goods have grown to 3.3% . Over the past decade, counterfeit products have flooded Bangladesh's market. The National Consumers' Rights Protection Directorate identified counterfeit powdered milk and cosmetics in large stores in Banani, Gulshan, and other affluent areas during enforcement operations. During the raid, officers found that several facilities were making counterfeit consumer products without MRP, importer names, or addresses. Even during the COVID-19 pandemic, dishonest businessmen created fake masks, gloves, and sanitisers. A group of dishonest merchants has started selling phoney and low-quality goods during the coronavirus crisis. Famous national and international brands are being forged alarmingly. This includes food, medicine, and lifesaving items.

Blockchain technology presents a promising solution to the widespread problem of counterfeit products, which not only cause financial losses but also pose health and safety risks to consumers. Counterfeit goods negatively impact businesses by damaging brand reputation, causing revenue losses, and increasing costs associated with fraud prevention. Traditional authentication methods, such as QR codes and RFID tags, have limitations, including susceptibility to duplication and high processing power requirements. Blockchain, with its decentralized and tamper resistant ledger, offers a secure and transparent way to track products throughout the supply chain, ensuring their authenticity and preventing fraudulent activities. A blockchain-based system records a product’s manufacturing details, including its initial production date, expiry date, and unique identification code (QR code, barcode, or RFID). These details are encrypted and stored on the blockchain, making unauthorized modifications impossible. When a consumer scans a product’s code, the system verifies it against the blockchain database. If the code matches, the product is authenticated; otherwise, alerts are sent to both the consumer and

manufacturer about potential counterfeiting. To further strengthen security, Physically Unclonable Functions (PUFs) can be integrated with blockchain to address challenges associated with extracting scanned product details. One major challenge in implementing blockchain-based anti-counterfeiting solutions is the need for widespread adoption by all supply chain participants, including manufacturers, distributors, and retailers. Resistance to data sharing and variations in blockchain preferences among businesses can hinder adoption. Additionally, the cost of deploying and maintaining a blockchain network can be high, particularly for small and medium-sized businesses (SMBs). However, these costs can be offset by long-term savings in fraud prevention and supply chain efficiency. Another concern is human error in entering product details into the blockchain. While blockchain ensures data integrity once recorded, incorrect or manipulated entries at the initial stage could still lead to false verification results. To mitigate this risk, an immutable ledger can store transaction logs of fake IDs, allowing real-time tracking and reporting of counterfeit activities. If counterfeit IDs are detected, access to the responsible entity is blocked, preventing further fraudulent actions. To enhance security and user incentives, an advanced system can impose penalties on store owners who fail to remove expired products from their inventory within a specified timeframe. These penalties can be issued in cryptocurrency, and rewards can be provided to customers who report fraudulent activities, encouraging greater consumer participation in counterfeit prevention. By implementing an algorithm to replace missing QR code bits, the framework reduces customer wait time during verification. for scanning and blockchain retrieval. The framework's secure and decentralized nature allows multiple stakeholders to access verified product information without the risk of tampering.

Blockchain technology is a popular topic among intellectuals. Blockchain has emerged in the supply chain, pharmaceutical, and health sectors. No third-party participation is a benefit of decentralized blockchain design. Blockchain has several advantages and can be used in industries where manufacturers can create blockchain-based QR codes, and customers can obtain real products. Manufacturers can transfer product ownership to sellers using blockchain technology, and buyers can verify product legitimacy before buying. Blockchain technology secures digital asset ownership certificates and preserves their physical properties. Blockchain makes it difficult for an attacker to edit, change, or replicate this data. Considering these prospects, a system architecture is developed that uses blockchain technology to protect product ownership and lets customers pick between counterfeit and authentic Ethereum smart contract products. This technology lets users

scan products using blockchain-based QR codes to verify authenticity, and it reduces manufacturing costs. This project aims to provide a solution to the growing problem of product counterfeiting and enhance the transparency and trustworthiness of supply chains using blockchain technology.

LITERATURE SURVEY

1.Title: A comparative study: Blockchain technology utilization benefits, challenges, and functionalities"

Author: Ali, Omar, et al

Year: 2021

Publisher: IEEE Access 9

Description: It gives us a through explanation of blockchain technology. It provides comprehensive details regarding the blockchain's history, advantages, disadvantages, architecture, and security. Blockchain is a powerful toolbox — not a universal hammer. When multiple, partially mistrusting parties need shared, auditable records or when new token-driven business models are required, blockchain can deliver substantial value. However, it carries trade-offs (performance, privacy, governance complexity) that make traditional databases or hybrid approaches preferable in many cases. Effective adoption depends on carefully matching architecture to the use-case, managing off-chain/on-chain boundaries, and planning governance and compliance from day one.

2. Title: "Identifying Fake Product Through a QR Code Based Blockchain System"

Author: P.LokeshReddy, A.MuniKarthik, M.Pavitra, Mrs.M.S.Soundarya,M.E.,

Year: 2022

Publisher : IRJET

Description: - The production and marketing of counterfeit or counterfeit items and products exposes consumers to financial, health and safety risks. It also affects the business growth of companies and businesses through loss of revenue, bad products, losses and price changes, many names are forced to use the money for legal protection. Use camera scanners to identify counterfeit products and link the product's QR code or barcode to blockchain blocks. If the numbers on the work match the numbers in the library, a notification is sent to the customer informing them about the originality of the work. If the customer accepts the offer, a notification will also be sent to the manufacturer stating where the customer purchased the product, if the number on the product does not match the number in the information. This approach aims to reduce counterfeiting by ensuring that consumers do not rely on retailers to decide whether the

product is genuine or not. Product tracking using blockchain technology involves creating tamper-proof and immutable information about products from production to delivery.

3. Title: "QR based Fake Product Detection using Blockchain"

Author: Rivaa Vadher, Rishith Vadher

Year: 2024

Publisher: IJIRT

Description: The proliferation of counterfeit products poses a significant challenge to consumers and brand owners alike. Current software solutions for counterfeit detection lack the robustness and transparency needed to combat this pervasive issue. This project proposes a novel approach to address this problem by leveraging blockchain technology in conjunction with QR codes. Our software solution combines blockchain technology and QR codes to create a secure and transparent anti counterfeiting system. Each authentic product is assigned a unique QR code linked to a blockchain-based ledger. When consumers scan the QR code, they can instantly verify the product's authenticity and access detailed information about its origin, manufacturing, and distribution history. The immutable nature of blockchain ensures data integrity, making it nearly impossible for counterfeiters to manipulate or replicate. By integrating blockchain with QR codes, this software enhances consumer confidence and empowers them to make informed purchasing decisions. Furthermore, brand owners gain real-time insights into their supply chains, enabling them to identify and mitigate counterfeit production more effectively.

PROBLEM STATEMENT

The worldwide improvement of an item or innovation consistently accompanies hazard factors, for example, forging and duplication. Forging items can influence the organization's name and the client's wellbeing. Presently days discovery of phony item is the greatest test. Fake items are causing a significant impact on the organization and the client's wellbeing. Hence, item creators are confronting enormous misfortune. India and different nations are battling such fake and fake items. In the proposed framework, the framework produces QR codes utilizing Blockchain innovation. This innovation stores exchange records in blocks. These squares are secure and difficult to access and change the data from it. By utilizing a QR code we can recognize the fake item**.** Compatibility across different web browsers posed challenges, as the project relied upon specific features of metamask extension that behaved differently across browsers. Ensuring that the frontend operates efficiently and responds promptly to user interactions, even when interacting with the blockchain, was a concern. Slow loading times or lag in transactions can deter users. Implementing security measures in the frontend to protect user data and private keys was critical. Failing to do so could expose users to risks like unauthorized access and data breaches.

OBJECTIVES

1. Ensure Product Authenticity

To verify whether a product is original or fake by scanning a unique QR code linked to tamper-proof blockchain records.

2. Establish a Transparent Supply Chain

To record every stage of the product lifecycle (manufacturing, packaging, shipping, retailing) on a blockchain so all stakeholders have a transparent view.

3. Provide Tamper-Proof Data Storage

To store product information (batch no., manufacturing date, serial details, ownership) on blockchain, where data cannot be modified or deleted.

4. Enable Real-Time Verification

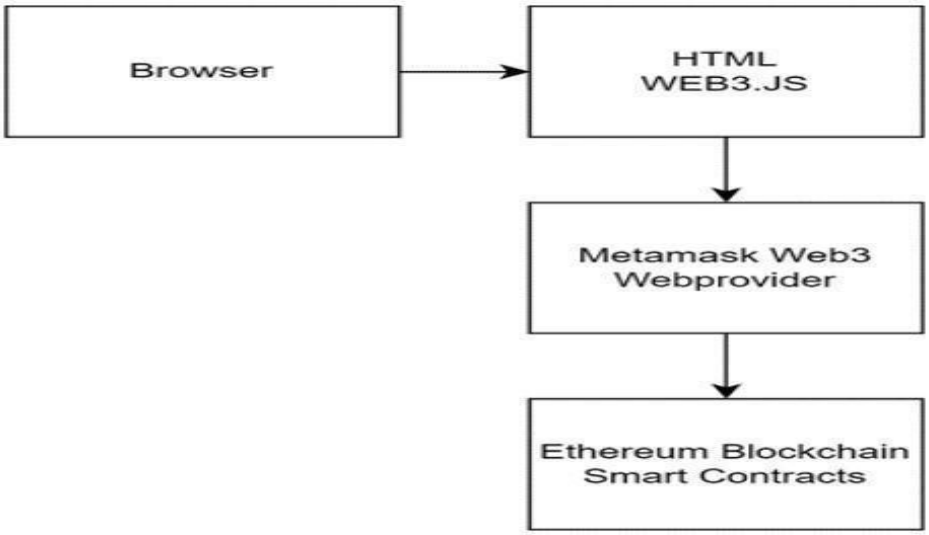
To allow customers, retailers, and authorities to instantly verify product authenticity using a mobile-based QR scanner.

5. Improve Trust Between Manufacturer & Consumers

To build customer confidence by providing a reliable method to check product genuineness before purchase.

PROPOSED METHODOLOGY

The proposed application system aims to combat the increasing prevalence of counterfeit goods by utilizing blockchain and QR codes. The system will store the product's source chain and maintain ownership histories, allowing buyers to access comprehensive product details and make informed purchasing decisions. QR codes will be used to validate products and add product information, while blockchain technology ensures the integrity of the stored data, preventing unauthorized alterations. By leveraging these technologies, the system will effectively identify counterfeit products, enhancing trust and transparency in the market place.



POSSIBLE OUTCOMES

1. Accurate Detection of Fake Products

- The system will successfully identify whether a product is **genuine or counterfeit** using a QR code.
- When a user scans the QR code, it will fetch real-time data from the blockchain and display authenticity status.

2. Immutable and Transparent Product History

- Every product's details (manufacture date, batch number, ownership history, location updates) will be **stored on blockchain**.
- No one can alter or delete these details, ensuring **trust and full transparency**.

3. Real-Time Verification for All Users

- Consumers can instantly verify any product before purchasing.
- Manufacturers and sellers can quickly view product status and history.

4. Prevention of Counterfeiting

- Since counterfeiters cannot manipulate blockchain data or generate identical QR codes:
    - Fake products will be easily detected.
    - Counterfeit activities will be reduced significantly.

5. Enhanced Trust Between Manufacturers & Consumers

- Buyers gain confidence because they can verify the product's origin.
- Manufacturers maintain brand reputation and reduce losses.

6. Secure Ownership Transfer

- Ownership transfers (Manufacturer → Seller → Consumer) are recorded securely on the blockchain.
- Creates a reliable supply chain traceability system.

7. Reduced Manual Errors

- No need for manual product verification.
- Automated QR verification reduces human mistakes and fraud opportunities.

8. Improved Supply Chain Visibility

- Manufacturers can: Track products across the entire supply chain. Identify unauthorized sellers.
- Sellers can: Check available products.
- Consumers can: View full product history.

9. Better Regulatory Compliance

- The system maintains **verifiable transaction logs**, making it easier for audits and legal compliance.
- Authorities can track fraud and take quick action.

10. User-Friendly Interface

- Easy to use dashboard for:
    - Manufacturers (Add products, Add sellers, Sell products)
    - Sellers (Sell to customers, Check availability)
    - Consumers (Verify products, Check history)

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Name and Signature of the Students

Name	USN	Signature
Disha D Pujar	2SR22CS012	
Kavya R Gyanagoudra	2SR22CS019	
Namratha V Kencharaddi	2SR22CS027	
Vanaja H Mallur	2SR22CS052	

Signature of Guide & Coordinator.

Dr. Murgesh V Jambigi **Ph.D**

Associate Prof., Dept. of CS&E

Signature of HOD

Prof. C. M. Parameshwarappa
M.Tech. (Soft. Engg)

Head of the Department, CS&E